Assignment 16 (ELEC 341 L16_FrequencyResponse)

Problem 1:

Convert the given transfer function into frequency response function, i.e., into $G(j\omega)$ form. Also, compute $M(\omega) = |G(j\omega)|$ and $\phi(\omega) = \angle G(j\omega)$. Find 20log $M(\omega)$ and $\phi(\omega)$ at $\omega = 20$.

$$G(s) = \frac{(s+3)(s+5)}{s(s+2)(s+4)} = \frac{s^2+8s+15}{s^3+6s^2+8s}.$$

$$G(jw) = \frac{(jw)^2+8jw+15}{(jw)^3+6(jw)^2+8jw} \qquad j^3 = j \cdot j^2 = -j$$

$$= \frac{-w^2+8jw+15}{-jw^3-6w^2+8jw} = \frac{15-w^2+8jw}{-6w^2+j(8w-w^3)}$$

$$G(jw) = \frac{15-w^2+8jw}{-6w^2+j(8w-w^3)}$$

$$M(\omega) = |G(j\omega)| = \frac{\sqrt{(15-w^2)^2+(8w)^2}}{\sqrt{(6w^2)^2+(8w-w^3)^2}}$$

$$M(20) = \frac{\sqrt{(15-20^2)^2+8(20)^2}}{\sqrt{(6(20)^2)^2+(8(20)-20^3)^2}} \simeq 0.0508497445458$$

$$\boxed{20\log(M(20)) = -25.8742\ dB}$$

$$\phi(\omega) = \angle G(jw)$$

$$= \tan^{-1}\left(\frac{8w}{15-w^2}\right) - \tan^{-1}\left(\frac{8w-w^3}{-6w^2}\right)$$

$$\phi(w) = \tan^{-1}\left(\frac{8w}{15-w^2}\right) - \tan^{-1}\left(\frac{8w-w^3}{6w^2}\right)$$

$$\boxed{\phi(20) = -95.5465^\circ}$$